Fiscal Year:	FY 2011	**Task Last Updated:**	FY 02/07/2011
PI Name:	Ferl, Robert Ph.D.		
Project Title:	Transgenic Plant Biomonitors of Space Flight Exposure		
Division Name:	Space Biology		
Program/Discipline:	SPACE BIOLOGY		
Program/Discipline--Element/Subdiscipline:	SPACE BIOLOGY--Cellular and molecular biology		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	(1) Cell & Molecular Biology (2) Plant Biology		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
PI Email:	robferl@ufl.edu	**Fax:**	FY 352-392-4072
PI Organization Type:	UNIVERSITY	**Phone:**	352-392-1928
Organization Name:	University of Florida		
PI Address 1:	Horticultural Sciences, 1301 Fifield Hall		
PI Address 2:	Box 110690		
PI Web Page:	http://www.hos.ufl.edu/		
City:	Gainesville	**State:**	FL
Zip Code:	32611-0690	**Congressional District:**	3
Comments:			
Project Type:	Flight	**Solicitation / Funding Source:**	98-HEDS-02
Start Date:	03/15/2007	**End Date:**	05/31/2012
No. of Post Docs:	2	**No. of PhD Degrees:**	1
No. of PhD Candidates:	0	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	1
No. of Bachelor's Candidates:	1	**Monitoring Center:**	NASA KSC
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:	Shuttle/ISS		
Flight Assignment:	STS-129; STS-130 ; STS-131 ; ISS NOTE: New end date is 5/31/2012, per PI (Ed., 2/3/2011) NOTE: Extended to 5/31/2011 per NSSC info; original end date 3/14/2010 (9/2009)		
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Paul, Anna-Lisa (Co-PI: University of Florida)		
Grant/Contract No.:	NNX07AH27G		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	This proposal is designed to be the next-generation of our previous flight project, PGIM-01, and investigates the use of reporter gene technology for the molecular analysis of development and stress in plants within space flight environments. We have designed a series of transgenic plants that carry sensor promoter - reporter gene constructs that are capable of monitoring a variety of environmental and developmental influences that might be altered in microgravity and general space flight environments. The first generation of these experiments with the GUS reporter have been tested for biological veracity on KC-135 parabolic flight missions, and flew as PGIM-01 in the Plant Growth Facility on STS-93. Results of these KC-135 missions and STS-93 were used in this proposal to (1) design a second generation of sensor promoters to refine our definition of the cellular biological impact of microgravity and space flight; (2) design a second generation of reporter gene constructs that include real-time and non- destructive observations of reporter gene activity; (3) design and conduct short and longer term flight experiments that begin a comprehensive analysis of molecular genetic changes induced by space flight; and (4) develop methods to integrate reporter gene activities with data from DNA chip analysis of genome expression. The long-term goal is an understanding of the space flight induced molecular changes in plant gene activity, an understanding that will lead to hardware and/or genetic engineering modifications that will mitigate the negative impacts of space flight on plant growth and development.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	An understanding of signal transduction in higher plants (and by inference, all eukaryotes) is central to basic molecular biology research.
Task Progress:	The first launch of the first leg of the TAGES experiment was on 11-16-2009, on STS-129. That was followed by two more launches; 02-08-2010 on STS-130, and 04-05-2010 on STS-131. The yearly report for 2010 primarily describes pre-flight and flight operations, plus operations associated with sample return and set-up for analyses. Data analyses will be included in the final report scheduled for next year.
Bibliography Type:	Description: (Last Updated: 11/27/2022)
Abstracts for Journals and Proceedings	Paul A-L, Toghranegar S, Amalfitano CA, Sarafan A Giongo A, Triplett E, Lee P, Berinstain A, Ferl RJ. "Planetary analogs as models for biologically relevant extraterrestrial exploration." NASA Lunar Science Institute annual meeting, Ames Research Center, CA, July 2010. NASA Lunar Science Institute annual meeting, Ames Research Center, CA, July 2010. , Jul-2010
Abstracts for Journals and Proceedings	Paul A-L, Zupanska A, Osterow DT, Zhang Y, Shanker S, Farmerie WG , Ferl RJ. "The impact of Spaceflight on Arabidopsis: Deep Sequencing and DNA Arrays as Collaborative Readouts of the Transcriptome of Arabidopsis Seedlings and Undifferentiated Cells in Space." Spaceflight - Plants and Microbes ... Flight and Ground Results. 26th Annual Meeting of the American Society of Gravitational and Space Biology, National Harbor, MD, November 2010. Program and abstracts. 26th Annual Meeting of the American Society of Gravitational and Space Biology, November 2010. p. 24. , Nov-2010
Abstracts for Journals and Proceedings	Ferl RJ, Levine H, Reed D, Spinale A, Murdoch T, Regan M, Cox D, Paul A-L. "Transgenic Plant Biomonitors of Spaceflight Exposure." Spaceflight - Plants and Microbes ... Flight and Ground Results. 26th Annual Meeting of the American Society of Gravitational and Space Biology, National Harbor, MD, November 2010. Program and abstracts. 26th Annual Meeting of the American Society of Gravitational and Space Biology, November 2010. p. 24. , Nov-2010
Articles in Peer-reviewed Journals	Ferl RJ, Paul A-L. "Lunar plant biology--A review of the Apollo era." Astrobiology. 2010 Apr;10(3):261-74. PMID: 20446867 , Apr-2010
Articles in Peer-reviewed Journals	Visscher AM, Paul A-L, Kirst M, Guy CL, Schuerger AC, Ferl RJ. "Growth performance and root transcriptome remodeling of Arabidopsis in response to Mars-like levels of magnesium sulfate." PLoS One. 2010 Aug 23;5(8):e12348. PMID: 20808807 , Aug-2010